

CHA NRF Client Function Guide

Used in Containerized Deployment

User Guide

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1 CHA Network Repository Function Client Introduction

1.1 Purpose and Scope

The purpose of this document is to describe the implementation (Network Repository Function client NF- Registrator) of the Charging (CHA) Component, representing the Charging Function (CHF).

CHF uses the services provided by the Network Repository Function (NRF) to Register, Update, and Deregister its entries in the NRF as part of 5G Core Network (5GC) in a Service-Based Architecture (SBA).

Note: In Container environment, CHA NF- Registrator monitors the other CHA Access pods for availability and based on that it registers or deregisters the Charging network function (CHF) in the 5GC NRF. It also sends heartbeats to the NRF as long as the CHF is registered. For more information on NF Registrator configuration, see [CHA NF Registrator Configurable Parameter List](#).



2 CHA Network Repository Function Client Description

CHA Network Repository Function (NRF) client supports the integration of CHA (Charging Function (CHF)) with 5G Network Repository Function (NRF) through Nnrf interface. The CHF registers its service locations on the NRF.

CHA

- Charging Function (CHF) is exposing two services, Converged Rate and Charge SBI, for more information see Converged Rate and Charge Capability Guide or Spending Limit Control SBI, for more information see Network Policy Control Capability Guide.
- CHA NRF Client, acts as a REST client front end to support Registration, Update, Heartbeat, and Deregistration of the CHF in the NRF, using the operations of the "Nnrf_NFManagement" service API.
- CHA NRF Client supports OAuth2 based Authorization of service request over Nchf using the operations of the "Nnrf_AccessToken" service API.

NRF

- Maintains a registry (NF profile) of available Network Function (NF) instances and their supported services
- Allows other NF instances to subscribe to, and get notified about, the registration in NRF of new NF instances of a given type.
- Supports service discovery function. It receives NF Discovery Requests from NF instances, and provides the information of the available NF instances fulfilling certain criteria (for example, supporting a given service).
- Exposes a "Token Endpoint" to which the Access Token Request service can be requested by NF Service Consumers over Nnrf_AccessToken service.Nnrf Interface.

Nnrf Interface

The Nnrf interface is a service-based interface used to register, update, or unregister NFs profile in Network Repository Function (NRF) as per 3GPP specifications.

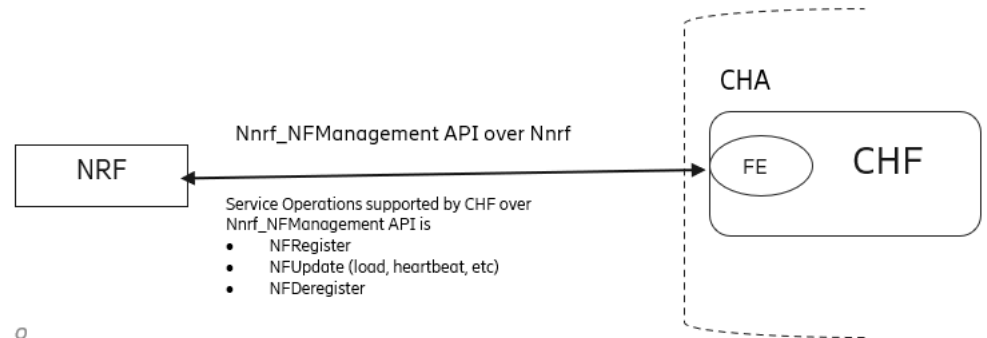


Figure 1 Overview of NRF and Ericsson Charging (CHF) Interaction for Nnrf_Management Service

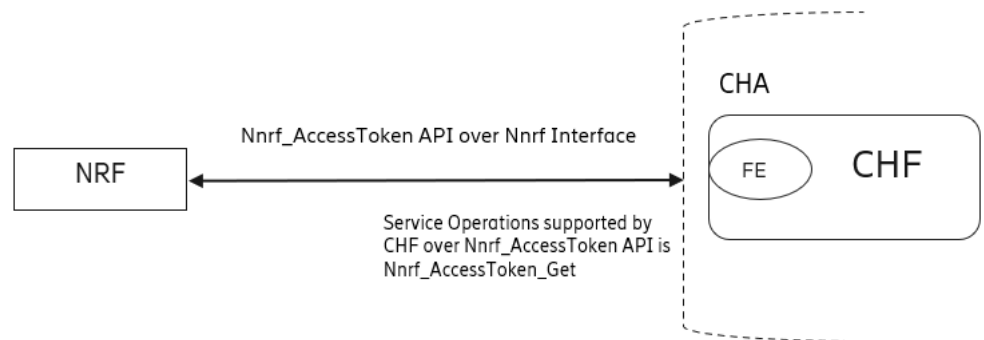


Figure 2 Overview of NRF and Ericsson Charging (CHF) Interaction for Nnrf_AccessToken Service

2.1 Configuration Management

2.1.1 NRF Configuration

To register CHF services in NRF and to get OAuth token from NRF OAuth2 server, it requires NRF server, and OAuth server details to communicate. NRF or OAuth server details like address, port are referred as NRF configuration.

BEAM CLI could be used for configuring the NRF server details in Containers. The NRF configuration commands provide the functionality to perform different operations towards the NRF configuration interface. The interface manages the configuration used by CHA for NRF connections. For more information, see [BEAM CLI Guide](#).



Note: When the OAuth feature 'FeatureFetchOAuthTokenForNnrfNfm' with Feature Control (FC) ID (fc39) is enabled, ensure that the required OAuth server configuration is completed to enable OAuth 2.0-based authorization when initiating communication with the NRF server. For more information on the OAuth server configuration, see [BEAM CLI Guide](#). Otherwise, NRF registration cannot be performed from CHA (CHF).

Table 1 NRF Server JSON Parameters

Parameter	Description
address	The NRF endpoint address. Format: IPv6 address should be given by enclosing within "[]" as mentioned in the format.http://<fqdn/ipv4/[ipv6]>:<port>
failureCodes	FailureCodes represents the list of HTTP result Code for which NRF failover happen (switch to next configured NRF). It can be any valid failure HTTP status code. Example: [404, 500].
secured	Indicates if TLS-based authentication is enabled.
compression	http:// Indicates if the requests to the NRF are going to be compressed.
nfProfileValidation	Enable support for NFProfile validation after registration.
nfServiceType	Indicates type of NFService where configurable values are nfservices, nfservicelist, or both.
appGrp	Application Group name default value is "global".

2.1.1.1 Payload Size Validation

The NRF Response is considered as failure if the response payload size exceeds maximum size of 16 MB.

2.1.2 NFProfile Configuration

The NF profile consists of general parameters of the CHF Instance, and the parameters of the two services exposed by the CHF Instance. It is used for registering the Charging Function (CHF) in the NRF.



NFProfile configuration enables to set and update the expected values as per the 3GPP spec in the CHF profile while registering and updating in NRF.

For more information, see BEAM CLI Guide.

Table 2 NFProfile for CHF Registration

Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
nfInstanceId	NfInstanceId	Mandatory	1	Unique identity of the NF Instance.	Individual NfInstanceId for each Registration	N
nfType	NFType	Mandatory	1	Type of Network Function	Configurable, Default: CHF	Y
nfStatus	NFStatus	Mandatory	1	Status of the NF Instance (NOTE 5)	REGISTERED	N
nfInstanceName	String	Optional	1	Human readable name of the NF Instance	Hostname of the CHA	N
heartBeatTimer	Integer	Mandatory		It can be included in the registration request. When present in the request it contains the heartbeat time proposed by the NF service consumer. This is	Configurable	Y



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				included in responses from NRF to registration requests (PUT) or in NF profile updates (PUT or PATCH). If the proposed heartbeat time is acceptable by the NRF based on the local configuration, it uses the same value as in the registration request; otherwise the NRF overrides the value using a preconfigured value.		
plmnList	array(PlmnId)	Conditional	1..N	PLMNs of the Network Function (NOTE 7). This	Configurable	Y



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				IE is present if this information is available for the NF.		
snpnList	array(PlmnIdNid)	Optional	1..N	SNPNs of the Network Function. This IE shall be present if the NF pertains to one or more SNPNs.	Configurable	Y
sNssais	array(ExtSnsai)	Optional	1..N	S-NSSAIs of the Network Function	Configurable	Y
perPlmnSnsaiList	array(PlmnSnsai)	Optional	1..N	This IE can be included when the list of S-NSSAIs supported by the NF for each PLMN it is supporting is different.	Configurable	Y
nsiList	array(string)	Optional	1..N	NSI identities of the Network Function	Configurable	Y



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
fqdn	Fqdn	Optional	0..1	FQDN of the Network Function (NOTE 1) (NOTE 2).	Applicable Note: The system determines the fqdn from the instance hostname.	N
interPlmnFqdn	Fqdn	Optional	0..1	If the NF needs to be discoverable by other NFs in a different PLMN, then an FQDN that is used for inter PLMN routing as specified in 3GPP TS 23.003 is registered with the NRF (NOTE 8). A change of this attribute results in triggering a "NF_PROFILE_CHANGED"	Not Applicable	



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				notification from NRF towards subscribing NFs located in a different PLMN, but the new value is notified as a change of the "fqdn" attribute.		
ipv4Addresses	array(Ipv4Addr)	Conditional	1..N	IPv4 addresses of the Network Function (NOTE 1) (NOTE 2)	Applicable	N
ipv6Addresses	array(Ipv6Addr)	Conditional	1..N	IPv6 addresses of the Network Function (NOTE 1) (NOTE 2)	Applicable	N
allowedPLMNs	array(PLMNid)	Optional	1..N	PLMNs allowed to access the NF instance.	Configurable	Y
allowedSNPNs	array(PLMNidNid)	Optional	1..N	SNPNs allowed to access the NF instance.	Configurable	Y
allowedNFTypes	array(NFType)	Optional	1..N	Type of the NFs	Configurable.	Y



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				allowed to access the NF instance.	Default Values: "SMF, SMSF, PCF"	
allowedNFDomains	array(string)	Optional	1..N	Pattern (regular expression according to the ECMA-262 dialect [8]) representing the NF domain names allowed to access the NF instance.	Configurable	Y
allowedNSSAIs	array(ExtNssai)	Optional	1..N	S-NSSAI of the allowed slices to access the NF instance.	Configurable	Y
priority	integer	Optional	0..1	Priority (relative to other NFs of the same type) in the range of 0-65535, to be used for NF selection; lower values indicate	Configurable	Y



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				a higher priority. The NRF can overwrite the received priority value when exposing an NFProfile with the Nnrf_NF Discovery service.		
capacity	integer	Optional	0..1	Static capacity information in the range of 0-65535, expressed as a weight relative to other NF instances of the same type.	Configurable	Y
load	integer	Optional	0..1	Indicates overload on the NF. 100 is sent to NRF when there is overload, 0 is sent only when no overload and 100	Not Configurable. System populate the value only during the load reporting and registering with NRF. If	N



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				is sent earlier.	no overload, then 0 is sent while registering NF profile	
loadTime Stamp	DateTime	Optional	0..1	Indicates the point in time in which the latest load information is generated at the NF Instance.	Not Configurable. System populates the value while sending the load information towards NRF.	N
locality	String	Optional	0..1	Operator defined information about the location of the NF instance (for example, geographic location, data center) (NOTE 3)	System populates based on the site configuration.	
udrInfo	UdrInfo	Optional	0..1	Specific data for the UDR (ranges of SUPI, group ID ...)	Not Applicable	



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
udrInfoList	map(UdrInfo)	Optional	1..N	Multiple entries of UdrInfo	Not Applicable	
udmInfo	UdmInfo	Optional	0..1	Specific data for the UDM (ranges of SUPI, group ID...)	Not Applicable	
udmInfoList	map(UdmInfo)	Optional	1..N	Multiple entries of UdmInfo	Not Applicable	
ausfInfo	AusfInfo	Optional	0..1	Specific data for the AUSF (ranges of SUPI, group ID...)	Not Applicable	
ausfInfoList	map(AusfInfo)	Optional	1..N	Multiple entries of AusfInfo	Not Applicable	
amfInfo	AmfInfo	Optional	0..1	Specific data for the AMF (AMF Set ID, ...)	Not Applicable	
amfInfoList	map(AmfInfo)	Optional	1..N	Multiple entries of AmfInfo	Not Applicable	
smfInfo	SmfInfo	Optional	0..1	Specific data for the SMF (DNN's, ...)	Not Applicable	
smfInfoList	map(SmfInfo)	Optional	1..N	Multiple entries of SmfInfo	Not Applicable	
upfInfo	UpfInfo	Optional	0..1	Specific data for the UPF	Not Applicable	



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				(S-NSSAI, DNN, SMF serving area, interface ...)		
upfInfoList	map(UpfInfo)	Optional	1..N	Multiple entries of UpfInfo	Not Applicable	
pcfInfo	PcfInfo	Optional	0..1	Specific data for the PCF	Not Applicable	
pcfInfoList	map(PcfInfo)	Optional	1..N	Multiple entries of PcfInfo	Not Applicable	
bsfInfo	BsfInfo	Optional	0..1	Specific data for the BSF	Not Applicable	
bsfInfoList	map(BsfInfo)	Optional	1..N	Multiple entries of BsfInfo	Not Applicable	
chfInfo	ChfInfo	Optional	0..1	Specific data for the CHF	Configurable	Y
chfInfoList	map(ChfInfo)	Optional	1..N	Multiple entries of ChfInfo	Configurable	Y
nefInfo	NefInfo	O	0..1	Specific data for the NEF	Not Applicable	
nrfInfo	NrfInfo	Optional	0..1	Specific data for the NRF	Not Applicable	
udsfInfo	UdsfInfo	O	0..1	Specific data for the UDSF	Not Applicable	
udsfInfoList	map(UdsfInfo)	O	1..N	Multiple entries of udsfInfo.	Not Applicable	



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
nwdafInfo	NwdafInfo	O	0..1	Specific data for the NWDAF.	Not Applicable	
pcscfInfoList	map(PcscfInfo)	O	1..N	Specific data for the P-CSCF.	Not Applicable	
hssInfoList	map(HssInfo)	O	1..N	Specific data for the HSS.	Not Applicable	
customInfo	object	Optional	0..1	Specific data for custom Network Functions	Configurable; User-Defined Structure, values	Y
recoveryTime	DateTime	Optional	0..1	Time stamp when the NF was (re)started (NOTE 5) (NOTE 6)	Not Applicable	
nfServicePersistence	boolean	Optional	0..1	If present, and set to true, it indicates that the different service instances of a same NF Service in this NF instance, supporting a same API version, are capable	Not Supported	



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				to persist their resource state in shared storage and therefore these resources are available after a new NF service instance supporting the same API version is selected by an NF Service Consumer (see 3GPP TS 23.527). Otherwise, it indicates that the NF Service Instance s of a same NF Service are not capable to share resource state inside the NF Instance.		
nfService s	array(NF Service)	Optional	1..N	List of NF Service	Charging, Spending	



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				Instances. It includes the services produced by the NF that can be discovered by other NFs, if any. (Note 8)	Limit Control Service Profile of individual member	
nfService List	map(NF service)	Optional	1..N	Map of NF Service Instances, where the "serviceInstanceId" attribute of the NFService object shall be used as the key of the map. It shall include the services produced by the NF that can be discovered by other NFs, if any. (Note 8)	Charging, Spending Limit Control Service Profile of individual member	



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
nfProfileChangesSupportInd	boolean	Optional	0..1	NF Profile Changes Support Indicator.	Not Supported	
nfProfileChangesInd	boolean	Optional	0..1	NF Profile Changes Indicator.	Not Supported	
defaultNotificationSubscriptions	array(DefaultNotificationSubscription)	Optional	1..N	Notification endpoints for different notification types	Not Supported	
lmfInfo	LmfInfo	Optional	0..1	Specific data for the LMF	Not Supported	
gmlcInfo	GmlcInfo	Optional	0..1	Specific data for the GMLC	Not Supported	
nfSetIdList	array(NfSetId)	Optional	1..N	NF Set ID defined in clause 28.12 of 3GPP TS 23.003 [12]. At most one NF Set ID shall be indicated per PLMN of the NF. This information shall be present if available.	Configurable	Y



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
servingScope	array(string)	Optional	1..N	The served areas of the NF instance. The absence of this attribute does not imply that the NF instance can serve every area in the PLMN.	Configurable	Y
lcHSupportInd	boolean	Optional	0..1	This IE indicates whether the NF supports Load Control based on LCI Header (see clause 6.3 of 3GPP TS 29.500). - true: the NF supports the feature. - false (default): the NF does not support the feature.	Not Supported	



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
olcHSupportInd	boolean	Optional	0..1	This IE indicates whether the NF supports Overload Control based on OCI Header (see clause 6.4 of 3GPP TS 29.500). - true: the NF supports the feature. - false (default): the NF does not support the feature.	Configurable	N
nfSetRecoveryTimeList	map(DateTime)	Optional	1..N	Map of recovery time, where the key of the map is the NfSetId of NF Sets that the NF instance belongs to. When present, the value of each entry of the map	Not Supported	



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				shall be the recovery time of the NF Set indicated by the key.		
serviceSetRecoveryTimeList	map(DateTime)	Optional	1..N	Map of recovery time, where the key of the map is the NfServiceSetId of the NF Service Sets configured in the NF instance. When present, the value of each entry of the map shall be the recovery time of the NF Service Set indicated by the key.	Not Supported	
scpDomains	array(string)	Optional	1..N	When present, this IE shall carry the	Configurable	Y



Attribute Name	Data Type	P	Cardinality	Description	CHF Profile	System Configuration (Y/N)
				list of SCP domains the SCP belongs to, or the SCP domain the NF (other than SCP) belongs to.		
scpInfo	ScpInfo	Optional	0..1	Specific data for the SCP	Not Applicable	

For more information, see NFProfile Configuration.

2.2 NF Management

The definitions for the Nnrf_NFManagement Service API are based on 3GPP TS 29.510 V15.6.0 (2019-03). This Function guide covers the Nnrf_NFManagement Service API parts that are applicable for the CHF.

2.2.1 API URI

URIs of this API have the following root:

{apiRoot}/{apiName}/{apiVersion}/...

where "apiRoot" is defined in subclause 4.4.1 of 3GPP TS 29.501, the "apiName" is set to "nnrf-nfm" and the "apiVersion" is set to "v1" for the current version of this specification.

2.2.1.1 Resource-URI Structure

Resource-URI of NRF SBI-request have the following structure:

{apiRoot}/{apiName}/{apiVersion}/nf-instances/{nfInstanceID}

{apiRoot}:

It would be combination of.



- Scheme (http://, https://)
- Authority (host and optional-port details) as defined in RFC 3986
- An optional deployment-specific string that starts with a "/" character.

{apiName}: nnrf-nfm

{apiVersion}: v1

nf-instances: static string

{nfInstanceID}: UUID of registering NF

Example: http://network-nrf.operator.com/nnrf-nfm/v1/nf-instances/a1af5084-9cc5-49c6-9876-123456789012

2.2.2 Location Header

As part of HTTP-‘201 Created’ response, NRF sends its actual URI (that is, FQDN in URI can be resolved to IP address) in the Location header parameter, if received, it is used as URI for further NRF message handling that is, NFUpdate/NFDeregister towards NRF.

Note: URI received in Location header is of the same format as indicated in example: http://150.236.1.100:31380/nnrf-nfm/v1/nf-instances/a1af5084-9cc5-49c6-9876-123456789012

2.2.3 Usage of HTTP

2.2.3.1 General

HTTP/2, as defined in IETF RFC 7540, is used as specified in clause 5 of 3GPP TS 29.500.

HTTP/2 is transported as specified in subclause 5.3 of 3GPP TS 29.500.

HTTP messages and bodies for the Nnrf_NFManagement service complies with the OpenAPI specification contained in Annex A of 3GPP TS 29.510.

2.2.3.2 HTTP Standard Headers

Support for HTTP Request or Response headers is provided as per the tables.



Table 3 HTTP Request Headers

Request Header	Populated by CHF Service	Mandatory or Optional	Value Supported	Comments
Accept	Yes	Optional	application/json and application/problem+json	
Accept-Encoding	Yes	Optional	gzip or identity	This header is populated based on the configuration parameter "compression". If "compression" is enabled, then "gzip" is populated otherwise "identity" is populated.
Content-Length	Yes	Mandatory		
Content-Type*	Yes	Mandatory Optional (For DELETE)	application/json or application/json-patch+json	
User agent	Yes	Optional		
Cache-Control	No	Mandatory	no-cache, no-store, no-transform, must-revalidate	
If-Modified-Since	No	Optional		
If-None-Match	No	Optional		
If-Match	No	Optional		
Via	No	Optional		
Authorization	Function Controlled	Optional	Access token fetched via Nnrf_AccessToken service can be populated	



Request Header	Populated by CHF Service	Mandatory or Optional	Value Supported	Comments
			when "5GAccessTokenClaimsValidation" function is enabled.	
3gpp-Sbi-Message-Priority	Yes	Optional		
Content-Encoding	Yes	Optional	gzip or identity	This header is populated based on the configuration parameter "compression". If "compression" is enabled, then "gzip" is populated otherwise "identity" is populated.
3gpp-Sbi-Sender-Timestamp	Yes	Optional		This header contains the date and time (with a millisecond granularity) at which an HTTP request is originated.

Table 4 HTTP Response Headers

Response Header	Processed by CHF Service	Mandatory or Optional	Value Supported	Comments
Content-Length	Yes	Optional(Note -1)		For integrity check with received content, Content-Length value is used.
Content-Type*	Yes	Yes	"application/json" or	If not received,

Response Header	Processed by CHF Service	Mandatory or Optional	Value Supported	Comments
			"application/json-patch+json" for a successful response or "application/problem+json" for a failure response.	Content is processed assuming it as "application/json" or "application/problem+json" format.
Location	Yes	Optional	Shall contain the URI of the created resource	Sent only in Response by NRF for NFRegister request
Retry-After	No	Optional		
Content-Encoding	No	Optional		
Cache-Control	No	Optional		
Age	No	N/A		
Last-Modified	No	N/A		
ETag	No	N/A		
Via	No	Optional		
3gpp-Sbi-Message-Priority	Yes	Optional		
WWW-Authenticate	No	Optional		Not validated by CHF

Note: Except for HTTP-Response 204 "No Content", Content-Length is mandatory.

Even though few HTTP headers are specified as Mandatory by 3GPP specifications, it signifies Network Function must support receiving this header but not mandate that header so many headers are considered Optional.

The following content types are supported:

- • JSON, as defined in IETF RFC 8259, is used as content type of the HTTP bodies specified in the present specification as indicated in subclause 5.4 of 3GPP TS 29.500.



- • The Problem Details JSON Object (IETF RFC 7807). The use of the Problem Details JSON object in an HTTP response body is signaled by the content type "application/problem+json".
- • JSON Patch (IETF RFC 6902). The use of the JSON Patch format in an HTTP request body is signaled by the content type "application/json-patch+json".

2.2.4 Nnrf_NFManagement Service Operations

2.2.4.1 Introduction

The services operations defined for the Nnrf_NFManagement service and that are applicable for CHF/Ericsson Charging are as follows:

- • NFRegister: It allows CHF's NF Instance to register its NF profile in the NRF; it includes the registration of the general parameters of the CHF's NF Instance, together with the list of services exposed by the CHF's NF Instance.
- • NFUpdate: It allows CHF's NF Instance to replace or update partially, the parameters of its NF profile (including the parameters of the associated services) in the NRF; it also allows to add or delete individual services offered by the CHF's NF Instance.
- • NFDeregister: It allows CHF's NF Instance to deregister its NF profile in the NRF, including the services offered by the CHF's NF Instance.

2.2.5 Registration

Registration with NRF can be done while CHA Access instance comes up and CHF 5Gc SBI services are available for consuming respective SBI requests.

The CHF sends a PUT request to the resource URI representing the NF Instance. The URI is determined by the NF Instance. The variable {nfInstanceID} represents an identifier, provided by the CHF. The format of the CHF's NF Instance ID is a Universally Unique Identifier (UUID) version 4, as described in IETF RFC 4122.

EXAMPLE: UUID version 4: "4947a69a-f61b-4bc1-b9da-47c9c5d14b64"

MULTIDRAG

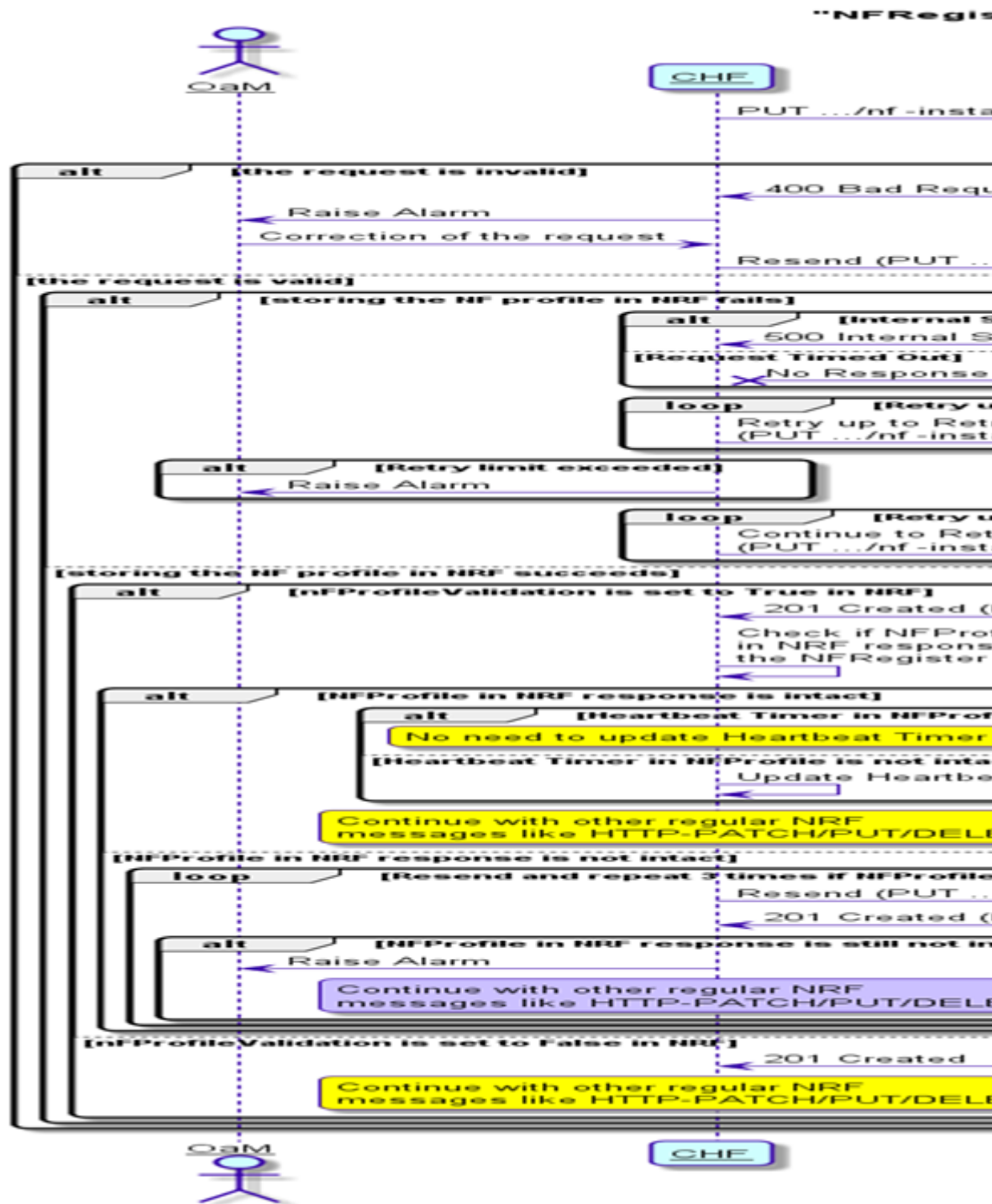


Figure 3 NRegister Operation Flow

- CHF's NF instance sends a PUT request to the NRF and includes the nfInstanceID and NFProfile



- • NRF checks the request, including the following aspects:
 - The profile in request body matches the schema of openapi.
 - The nfInstance from URI is identical with the one included in request body.
 - The profile in request body matches the rules defined in 3GPP TS 29.510. Such as SupiRange, either the "start" and "end" attributes, or the "pattern" attribute, is present.
- • If the request is invalid, error code "400 Bad Request" is returned indicating the problem. To this CHF raises an alarm.
- When the request is corrected, CHF resends the PUT request to the NRF
- If the request is valid, NRF attempts to store the NF Profile.
 - If storing the NF Profile fails, response is returned with error code "500 Internal Server Error".
 - If the NRF is not responding due to any reason, the request to NRF can be timed out.
 - To both the previous cases, CHF keeps retrying the sending of PUT request to the NRF until the retry limit internally configured in CHF is reached.
 - If the retry limit is exceeded and still the storing of NF profile fails, CHF raises an alarm.
 - CHF then keeps on retrying the request towards NRF until it succeeds.
- • If storing of the NF Profile succeeds and if nFProfileValidation is set to True in NRF, the NRF response is returned with status code "201 Created" and NFProfile including the heartBeatTimer. (See "NF Heartbeat" for more details on heartBeatTimer).
- • CHF compares the NFProfile received from NRF in the response and compares it with the NFProfile that it sent to NRF in the NfRegister request.
- If the NFProfile in NRF response is intact including the heartBeatTimer, CHF does not do any change and continues further communication with NRF (HTTP-PATCH/PUT/DELETE) using the Location header URI from the NRF response.
- If the NFProfile in NRF response is intact except the heartBeatTimer, CHF updates the heartBeatTimer value based on the NFProfile received in the NRF response change and continues further communication with NRF (HTTP-PATCH/PUT/DELETE) using the Location header URI from the NRF response.

- If the NFProfile in NRF response is NOT intact at all, CHF resends the PUT request to the NRF including the nfInstanceID, and NFProfile and compares the NFProfile in NRF response with the NFProfile sent in the request.
- If the NFProfile in NRF response is not intact, CHF repeats sending of the PUT request 2 more times (total three times). And if the NFProfile is still not intact, CHF raises an alarm and continues further communication with NRF (HTTP-PATCH/PUT/DELETE) using the Location header URI from the NRF response.
- If storing of the NF Profile succeeds and if nFProfileValidation is set to False in NRF, the NRF response is returned only with status code "201 Created" and no NFProfile. CHF continues further communication with NRF (HTTP-PATCH/PUT/DELETE) using the Location header URI from the NRF response.
- The load and loadTimeStamp are also sent as part of NF profile during NF registration.

2.2.6 Deregistration

This service operation removes the profile of a CHF's NF instance previously registered in the NRF.

It is executed by deleting a given resource identified by an "NF Instance ID". The operation is started by issuing a DELETE request on the URI representing the CHF's specific NF Instance.

Note: NFDeregister happens in two scenarios:

- when no CHA access pod instances are running.
- or -
- when all the NRF endpoints are removed from the configurable list

The NFDeregister message will be sent only to the NRF endpoint where the last successful interaction happened

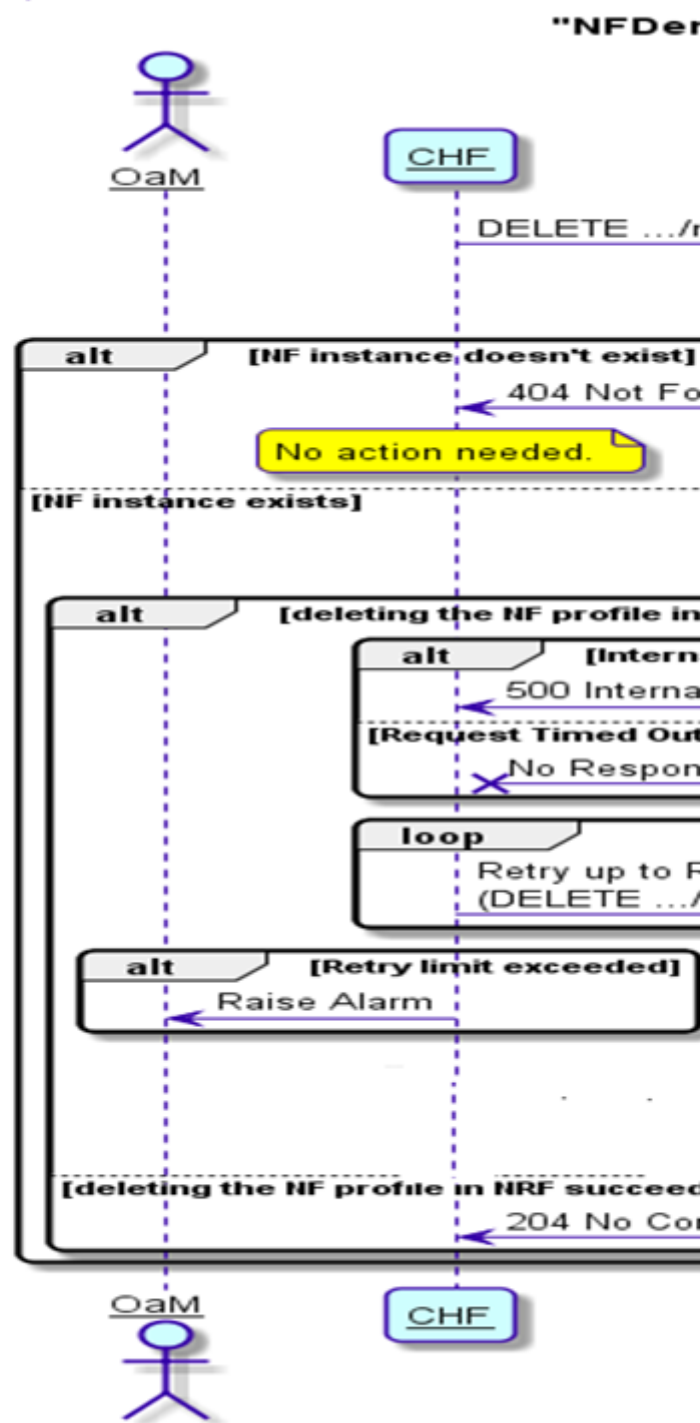


Figure 4 NFDeregister Operation Flow

- CHF sends a DELETE request to the NRF and includes the nfInstanceID of the NF Instance that needs to be deregistered.

- NRF retrieves this CHF's NF instance by instance ID.
- If this NF instance ID does not exist, the request is rejected with error code "404 Not Found". CHF not take any further action to this.
- If the NF Instance ID exists, NRF removes the profile of this NF instance.
 - If deleting the profile fails because of internal NRF problem, response is returned with error code.
 - If the NRF is not responding because of any reason, the request to NRF be timed out.
 - To both the previous cases, CHF keeps retrying the sending of DELETE request to the NRF until the retry limit configured in CHF (CHA) is reached.
 - If the retry limit is exceeded and still the deletion of the NF profile fails, CHF raises an alarm.
- If deleting the profile succeeds, response is returned with status code "204 No Content".

2.2.7 Update

This service operation updates the profile of a Network Function (CHF) previously registered in the NRF by providing the updated NF profile of the requesting CHF to the NRF. This service operation is triggered when there is a configuration change.

The update operation can apply to the whole profile of the CHF (complete replacement of the existing profile by a new profile) or it can apply only to a subset of the parameters of the CHF profile (including adding/deleting/replacing services to the CHF profile).

2.2.7.1 Update with HTTP PUT Request

NFUpdate with HTTP PUT request is sent by CHF towards NRF to perform a complete replacement of the NFProfile of the CHF Network Function Instance.

The load and loadTimeStamp are also sent as part of NF profile during NF update.

2.2.7.2 Update with HTTP PATCH Request

When there is a partial update of the NFProfile for an existing CHF/CHF's Network Function Instance, the CHF sent an HTTP PATCH request with NFUpdate. This allows the addition, removal, or replacement of one or more parameters within the CHF/CHF's of Network Function Instance.



The NFProfile Partial Update aligns with RFC 6902, which specifies the JSON Patch format for expressing partial updates to JSON documents. However, it is important to note that the network function Profile Partial Update specifically supports the replacement of a list of objects as a whole, rather than modifying an individual entries within the list. Only exception to it is "nfServices" parameter.

Existing NFProfile:

```
{
  "nfInstanceId": "d28c5425-e2c1-4223-b84d-493bc38bf11f",
  "nfInstanceName": "214.5.231.197",
  "nfType": "CHF",
  "nfStatus": "REGISTERED",
  "heartBeatTimer": 120,
  "fqdn": "seliicvm03958.bss-te.seli.gic.ericsson.se",
  "ipv4Addresses": [
    "214.5.231.197"
  ],
  "ipv6Addresses": [
    "fc00:1000::f38b"
  ],
  "allowedNfTypes": [
    "SMF",
    "SMSF",
    "PCF"
  ],
  "priority": 0,
  "loadTimeStamp": "2024-05-18T13:38:51.000+02:00",
  "load": 0,
  "locality": "Karlskrona",
  "nfServicePersistence": false,
  "nfServices": [
    {
      "serviceInstanceId": "nchf-convergedcharging",
      "serviceName": "nchf-convergedcharging",
      "versions": [
        {
          "apiVersionInUri": "v2",
          "apiFullVersion": "2.0.3"
        },
        {
          "apiVersionInUri": "v3",
          "apiFullVersion": "3.0.1"
        }
      ],
      "allowedNssais": [
        {
          "sst": 210,
          "sd": "223",
          "sdRanges": [
            {
              "start": "230",
              "end": "290"
            }
          ]
        },
        {
          "wildcardSd": true
        }
      ],
      "scheme": "https",
      "nfServiceStatus": "REGISTERED",
      "fqdn": "seliicvm03958.bss-te.seli.gic.ericsson.se",
      "ipEndpoints": [
        {
          "ipv4Address": "214.5.231.197",
          "transport": "TCP",
          "port": 13444
        },
        {
          "ipv6Address": "fc00:1000::f38b",
          "transport": "TCP",
          "port": 13444
        }
      ],
      "supportedFeatures": "2e",
      "oauth2Required": true,
    }
  ]
}
```



```
        "allowedNfTypes": [
            "SMF",
            "SMSF"
        ],
    },
    {
        "serviceInstanceId": "nchf-spendinglimitcontrol",
        "serviceName": "nchf-spendinglimitcontrol",
        "versions": [
            {
                "apiVersionInUri": "v1",
                "apiFullVersion": "1.1.1"
            }
        ],
        "scheme": "https",
        "nfServiceStatus": "REGISTERED",
        "fqdn": "seliicvm03958.bss-te.seli.gic.ericsson.se",
        "ipEndPoints": [
            {
                "ipv4Address": "214.5.231.197",
                "transport": "TCP",
                "port": 13444
            },
            {
                "ipv6Address": "fc00:1000::f38b",
                "transport": "TCP",
                "port": 13444
            }
        ],
        "supportedFeatures": "3",
        "oauth2Required": true,
        "allowedNfTypes": [
            "PCF"
        ],
        "supportedVendorSpecificFeatures": {
            "000193": [
                {
                    "featureName": "ERICSSON_SLC",
                    "featureVersion": "1.0.0"
                }
            ]
        }
    }
],
"nfServiceList": {
    "nchf-spendinglimitcontrol": {
        "serviceInstanceId": "nchf-spendinglimitcontrol",
        "serviceName": "nchf-spendinglimitcontrol",
        "versions": [
            {
                "apiVersionInUri": "v1",
                "apiFullVersion": "1.1.1"
            }
        ],
        "scheme": "https",
        "nfServiceStatus": "REGISTERED",
        "fqdn": "seliicvm03958.bss-te.seli.gic.ericsson.se",
        "ipEndPoints": [
            {
                "ipv4Address": "214.5.231.197",
                "transport": "TCP",
                "port": 13444
            },
            {
                "ipv6Address": "fc00:1000::f38b",
                "transport": "TCP",
                "port": 13444
            }
        ],
        "supportedFeatures": "3",
        "oauth2Required": true,
        "allowedNfTypes": [
            "PCF"
        ],
        "supportedVendorSpecificFeatures": {
            "000193": [
                {
                    "featureName": "ERICSSON_SLC",
                    "featureVersion": "1.0.0"
                }
            ]
        }
    }
}
```



```

    ]
  },
  "nchf-convergedcharging": {
    "serviceInstanceId": "nchf-convergedcharging",
    "serviceName": "nchf-convergedcharging",
    "versions": [
      {
        "apiVersionInUri": "v2",
        "apiFullVersion": "2.0.3"
      },
      {
        "apiVersionInUri": "v3",
        "apiFullVersion": "3.0.1"
      }
    ],
    "scheme": "https",
    "nfServiceStatus": "REGISTERED",
    "fqdn": "seliicvm03958.bss-te.seli.gic.ericsson.se",
    "sNssais": [
      {
        "sst": 210,
        "sd": "223",
        "sdRanges": [
          {
            "start": "230",
            "end": "290"
          }
        ],
        "wildcardSd": false
      }
    ],
    "ipEndPoints": [
      {
        "ipv4Address": "214.5.231.197",
        "transport": "TCP",
        "port": 13444
      },
      {
        "ipv6Address": "fc00:1000::f38b",
        "transport": "TCP",
        "port": 13444
      }
    ],
    "supportedFeatures": "2e",
    "oauth2Required": true,
    "allowedNfTypes": [
      "SMF",
      "SMSF"
    ]
  }
}

```

Add parameters in NfProfile:

```

"interPlmnFqdn": "seliicvm03999.bss-te.seli.gic.ericsson.se",
"chfInfo": {
  "groupId": "232",
  "supiRanges": [
    {
      "start": "230",
      "end": "330",
      "pattern": "string"
    }
  ],
  "gpsiRanges": [
    {
      "start": "230",
      "end": "330",
      "pattern": "string"
    }
  ],
  "externalGroupIdentifiersRanges": [
    {
      "start": "230",

```



```

        "end": "330",
        "pattern": "string"
      }
    ],
    "supportedDataSets": [
      "SUBSCRIPTION",
      "POLICY"
    ]
  }
}

```

Replace parameters in NfProfile:

```

"fqdn": "seliicvm04599.bss-te.seli.gic.ericsson.se"
And under NfServices:
"allowedNssais": [
  {
    "sst": 220,
    "sd": "333",
    "sdRanges": [
      {
        "start": "230",
        "end": "490"
      }
    ],
    "wildcardSd": false
  }
]

```

Remove parameters in NfProfile:

Remove sNssais under NfServiceList.

Patch message request payload sent to NRF:

```

PATCH .../nf-instances/d28c5425-e2c1-4223-b84d-493bc38bf11f
Content-Length: 326
Content-Type: application/json-patch+json
[
  {
    "op": "add",
    "path": "interPlmnFqdn",
    "value": "tmo.com"
  },
  {
    "op": "add",
    "path": "chfInfo/groupId",
    "value": "232"
  },
  {
    "op": "add",
    "path": "chfInfo/supiRanges",
    "value": [
      {
        "start": "230",
        "end": "330",
        "pattern": "string"
      }
    ]
  },
  {
    "op": "add",
    "path": "chfInfo/gpsiRanges",

```




```

    "value": [
      {
        "start": "230",
        "end": "330",
        "pattern": "string"
      }
    ],
    {
      "op": "add",
      "path": "CHFInfo/plmnRangeList",
      "value": [
        {
          "start": "500",
          "end": "800",
          "pattern": "string"
        }
      ]
    }
  ]
}
]

```

2.2.8 Heart-beat

Each CHF's NF instance that has previously registered in NRF contacts the NRF periodically (heart-beat), by starting the NFUpdate service operation, to show that the CHF's NF instance is still operative.

The time interval "heartBeatTimer" at which the NRF is contacted is deployment-specific, and it is returned by the NRF to the CHF as a result of a successful registration of the NF instance.

The parameter "heartBeatTimer" contains the number of seconds expected between two consecutive heart-beat messages from an NF Instance of the CHF to the NRF.

When the NRF detects that a given NF instance of the CHF has not updated its profile for a configurable amount of time (longer than the heart-beat interval), the NRF changes the status of the CHF's NF instance to SUSPENDED, and considers that the NF instance and its services can no longer be discovered by other NFs via the NFDiscovery service. The NRF notifies NFs subscribed to receiving notifications of changes of the NF Profile that the NF status of the NF instance has been changed to SUSPENDED.

If the NRF modifies the heart-beat interval value of a given NF instance currently registered (for example, as a result of an OA&M operation), it returns the new value to the registered CHF's NF instance in the response of the next periodic heart-beat interaction received from that CHF's NF instance and, until then, the NRF applies the heart-beat check procedure according to the original interval value.

The CHF sends a PATCH request to the resource URI representing its NF Instance. The payload body of the PATCH request contains a "replace" operation on the "nfStatus" attribute of the NF Profile of the CHF's NF Instance, and set it to the value "REGISTERED".

CHF can provide load information for the NF instance, particularly at the instance level. When CHA Access is overloaded, then CHF sends both the load and

loadTimeStamp towards NRF. The load value is reset to zero along with the loadTimeStamp once the overload situation is resolved.

Example:

```
PATCH .../nf-instances/4947a69a-f61b-4bc1-b9da-47c9c5d14b64
Content-Type: application/json-patch+json
[
  { "op": "replace", "path": "/nfStatus", "value": "REGISTERED" },
  { "op": "replace", "path": "/load", "value": "0" },
  { "op": "replace", "path": "/loadTimeStamp", "value": "2024-05-20T12:31:02.000+05:30" }
]
HTTP/2 204 No Content
Content-Location: .../nf-instances/4947a69a-f61b-4bc1-b9da-47c9c5d14b64
```

2.2.9

Overload Reporting

CHF can report overload ratio to NRF based on current state of system. A mean overload ratio is calculated based on overload ratio reported from Access instances belonging to a specific NRF instance. This value is then compared towards configurable thresholds for overload. Overload ratio is evaluated in intervals based on Evaluation Interval configured in CHA GUI. For more information on the Evaluation Interval configuration, see Overload Control section in CHA Converged Rate and Charge Function Guide or CHA Network Policy Control Function Guide. For more information on Overload Reporting (Get, Update, and Delete) configuration, see BEAM CLI Guide.

upwardsOverloadThreshold: Defines an upper threshold for when to report overload to NRF. When reaching or passing above threshold, CHF is considered in overload (100).

downwardsNormalLoadThreshold: Defines a lower threshold for when to report normal load to NRF. When going below threshold, CHF is considered in normal load (0).

Example:

If mean overload ratio $\geq$ upwardsOverloadThreshold report load = 100

If mean overload ratio $<$ downwardsNormalLoadThreshold report load = 0

Load is only reported to NRF if overload ratio changes, such that 0 to 100 or 100 to 0.



2.2.10 Handling Multiple NRF connections

CHA/CHF supports multiple NRF end points to achieve redundancy, however, CHA registers the NF information with only one NRF at a time. If the first NRF fails, CHF switches over to update the second NRF.

CHF follows a failover strategy which means, it continues to use the second NRF until it fails.

When switching to alternate NRF, CHF will always send NFUpdate request with full replacement towards new NRF. CHF accepts both "201 Created" or "200 OK" for the NFUpdate request. The assumption is that the multiple NRF end points are in sync.

Note: Lack of synchronization between NRF end points may result in incorrect operation state of CHF in NRFs .

Note: NFDeregister message will be sent only to NRF endpoint where the last successful interaction happened.

2.3 Security Management

The NRF offers an Nnrf_AccessToken service (used for OAuth2 authorization, see IETF RFC 6749 [9]), following the "Client Credentials" authorization grant, as specified in 3GPP TS 33.501 V15.4.0 [11]. It exposes a "Token Endpoint" where the Access Token Request service can be requested by NF Service Consumers.

The NRF shall send access token to the NF service consumer in the Nnrf_AccessToken_Get response operation by generating an access token with appropriate claims included and digitally signing the access token based on a shared secret or private keys as described in RFC 7515 [12]. Each access token has a validity time as specified in its expiration time included in the response.

The NF service consumer can store the received tokens. Stored tokens can be reused for accessing services from producer NF type listed in claims (scope, audience) during its validity time.

2.4 Fault Management

2.4.1 Retry Handling

In the cases where CHF sends NFRegister/NFUpdate/NFDeregister / AccessTokenRequest towards the NRF and if NRF returns failure HTTP status code like 500 "internal server error" to CHF OR if there is no response from NRF and the request from CHF to NRF over the Nnrf SBI interface times out, CHF uses the internal application configuration related to the following parameters for retrying the requests towards NRF in these cases.



- If error code returned from NRF matches with the value configured in failureCodes, then no retry is done and only alarm be raised.
- If error code returned from NRF does not matches with the value configured in failureCodes, then retry is done as per retriesForNrfConnection, retryIntervalForNrfConnection, connectionTimeout, and responseTimeout. CHA raises alarm once number of retries are done and request is still not successful.

In both the previous cases,

- if there is only one NRF configured in CHF, then towards same NRF, NFUpdate will be triggered post raising alarm after number of retries.
- Otherwise, alternative NRF is selected and NFUpdate operation be triggered.

For more information, see Handling multiple NRF connections.

Note: The retry handling is also applicable if there is connection failure towards an NRF instance.

Table 5 Application Config Parameters for Handling Retries

Attribute Name	Data Type	Description	Default Value
retriesForNrfConnection	Integer	Count of retrying the NRF connection	3
retryIntervalForNrfConnection	Integer	Time-out interval (in seconds) for retrying the Nrf connection	30
connectionTimeout	Integer	Time (with range 0 to 1000 seconds) to wait to establish a connection with NRF.	10
responseTimeout	Long	Time (in milliseconds) to wait for response from NRF.	1000
FailureCodes Note: This attribute should be configured when there are multiple NRF end points.	Integer	Represents the list of HTTP result Code for which OAuth server failover happen (switch to next configured OAuth server). It can be	None



Attribute Name	Data Type	Description	Default Value
		any valid failure HTTP status code.	

2.5 Performance Management

2.6 Alarms

There are two alarms that can be raised by CHF based on the cases where communication towards NRF cannot be established.

1. Major Alarm: This alarm is only raised in a case where CHF cannot establish OR loses communication towards an NRF server.

Table 6

Title: Could Not Establish Connection with at Least One NRF.	
Component or Module	CHA.Access (CHF)
Severity	MAJOR
Alarm Category	Communications Alarm
Probable cause	Connection Establishment Error (22)
Notification Type	RAISE_ALARM
Alarm description	Charging, no connection with Network Repository Function
Proposed action	Verify that NRF is up and running and that the instances can reach each other.

2. Critical Alarm: This alarm is raised in a case where there no communication to any NRF at all.

Table 7

Title: Could Not Establish Connection with Any NRF.	
Component or Module	CHA.Access (CHF)
Severity	CRITICAL
Alarm Category	Communications Alarm
Probable cause	Connection Establishment Error (22)
Notification Type	RAISE_ALARM
Alarm description	Charging, no connection with any Network Repository Function



Title: Could Not Establish Connection with Any NRF.	
Proposed action	Verify that NRF is up and running and that the instances can reach each other.

2.7 Result and Response Codes

2.7.1 Retry Handling When NRF Returns HTTP Status Code 400 and 404

404 is returned when there is no session/nfinstanceId exists in the NRF and 400 is returned when NRF fails because of errors in the encoding the NFProfile JSON object. The following process follows:

1. CHF raises an alarm.
2. CHF Retries "NFRegister" (NFUpdate) with the whole NFProfile to the same NRF for the configured number of times.
3. CHF continues retrying with the same NRF till it gets the same response Code.